

HEALTHCONNECT CLINIC

EXPERIENCE LAB

Week 6 – Advanced Analytics & Decision Support Report

Track: Data Analytics

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Tool: Microsoft Power BI

Dataset: HealthConnect Appointment Data

1. Week 5 → Week 6 Transition

During Week 5, the Data Analytics track prepared and analysed the HealthConnect appointment dataset and developed a Power BI dashboard to understand appointment attendance and no-show behaviour.

The main Week 5 output was an interactive Power BI dashboard containing key performance indicators and visualisations covering appointment outcomes, reminders, appointment characteristics, patient characteristics, booking lead time, distance and waiting time.

One of the most important findings from Week 5 was the high level of missed appointments, with the no-show rate at approximately **48.5%**.

A key limitation identified was that the initial analysis mainly described the patterns in the data. Therefore, Week 6 focused on deeper analysis and validation of the most important findings rather than simply recreating the Week 5 analysis.

The Week 6 objective was to investigate the factors associated with no-show behaviour, validate the key KPIs and improve the dashboard so that the findings could better support HealthConnect decision-making.

This approach follows the Week 6 requirement to build on Week 5 rather than repeat the previous analysis.

2. Advanced Analysis

The Week 6 analysis focused on identifying patterns that could help HealthConnect understand which appointments may have a higher likelihood of being missed.

2.1 No-Show Rate by Previous No-Show History

The analysis showed a clear relationship between previous no-show history and the current appointment outcome.

Previous No-Shows	No-Show Rate
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0	43.5%
1	53.5%
2	59.4%
3	67.9%
4	66.7%
5	100.0%

The pattern shows that appointments associated with patients who had previous missed appointments generally had higher current no-show rates.

The strongest and most reliable pattern is the increase from **43.5% for patients with no previous no-shows to 67.9% for those with three previous no-shows.**

The 100% rate for five previous no-shows should be interpreted cautiously because this group contains only a very small number of appointments.

Decision-support implication: Previous no-show history can be considered an important indicator when identifying appointments that may require additional engagement or reminder support.

2.2 No-Show Rate by Reminder Status

The analysis also compared appointments according to whether a reminder was sent.

Reminder Sent No-Show Rate

No	51.4%
Yes	47.4%

Appointments where a reminder was sent had a lower observed no-show rate than appointments where no reminder was sent.

The difference is approximately **4 percentage points**.

However, this is an observational analysis and does not establish that reminders directly caused the reduction in no-shows. Other characteristics may influence whether a reminder is sent and whether a patient attends.

Decision-support implication: HealthConnect should continue monitoring reminder activity and consider targeted reminder strategies for higher-risk appointments.

2.3 No-Show Rate by Appointment Type

No-show rates varied across appointment types.

Appointment Type	No-Show Rate
Follow-up	51.2%
Diagnostic Test	49.8%
Specialist Consultation	47.4%
General Consultation	46.7%

Follow-up appointments recorded the highest observed no-show rate, while General Consultations recorded the lowest.

Decision-support implication: Follow-up appointments may require additional investigation and targeted patient-engagement strategies.

2.4 No-Show Rate by Age Group

The analysis identified differences in no-show rates across age groups.

Age Group	No-Show Rate
55–64	50.8%
25–34	50.7%
18–24	50.2%
35–44	48.4%
45–54	48.1%
65+	45.1%

The highest observed rates were among the **55–64, 25–34 and 18–24** age groups, while the **65+** group had the lowest observed no-show rate.

These differences are useful for segmentation but should not be interpreted as evidence that age itself causes missed appointments.

2.5 Booking Lead Time

The Week 6 dashboard also examined the relationship between booking lead time and no-show rate.

The analysis indicated that no-show rates generally increased as the time between booking and the scheduled appointment became longer.

Decision-support implication: Appointments scheduled further in advance may benefit from additional reminders closer to the appointment date.

2.6 Distance and Waiting Time

Average distance to the clinic and waiting time were compared across appointment outcomes.

The dashboard showed relatively similar waiting times across attended, no-show and cancelled appointments, while average distance showed some variation.

This suggests that distance may warrant further investigation as an accessibility factor, while the current analysis does not provide strong evidence that waiting time alone explains no-show behaviour.

The dataset contains a small amount of missing distance and waiting-time information, so these findings should be treated as directional.

3. Validated Key Performance Indicators

The Week 5 KPIs were reviewed using the Week 6 analysis.

KPI	Validated Result
Total Appointments	5,000
No-Show Rate	48.5%
Attendance Rate	46.3%
Cancellation Rate	5.3%

The KPI results remained consistent with the Week 5 analysis.

The high no-show rate remains the most significant operational finding because almost half of the recorded appointments resulted in a no-show.

4. Key Decision-Support Findings

The deeper Week 6 analysis produced the following main findings:

Finding 1 – Previous no-show behaviour is an important risk indicator

No-show rates increased substantially as previous no-show history increased. This provides a useful basis for identifying appointments that may require additional engagement.

Finding 2 – Reminder activity is associated with lower observed no-show rates

Appointments with reminders had a lower observed no-show rate than appointments without reminders. However, this relationship should not be interpreted as causal.

Finding 3 – Some appointment types have higher no-show rates

Follow-up appointments had the highest observed no-show rate at approximately 51.2%.

Finding 4 – No-show behaviour varies across age groups

The analysis identified differences between age groups, suggesting that patient segmentation may be useful when designing engagement strategies.

Finding 5 – Longer booking lead times may increase the risk of missed appointments

The observed pattern suggests that appointments scheduled further ahead may require additional reminder activity closer to the appointment date.

5. Business Recommendations

Based on the Week 6 findings, the following actions are recommended for HealthConnect:

1. Introduce targeted engagement for patients with previous no-shows

Patients with a history of missed appointments should be considered for additional reminder or confirmation processes.

2. Strengthen reminder strategies

HealthConnect should monitor reminder coverage and consider sending reminders closer to appointment dates, particularly for appointments booked well in advance.

3. Review follow-up appointments

Because follow-up appointments showed the highest no-show rate, HealthConnect should investigate whether additional communication or confirmation is required for this appointment type.

4. Use patient segmentation

Age group and previous attendance history can be used to identify different patterns of appointment behaviour and support more targeted engagement.

5. Monitor appointment lead time

Longer booking lead times should be monitored because appointments scheduled further in advance may be more vulnerable to missed appointments.

6. Continue dashboard monitoring

The Power BI dashboard should be used to monitor no-show, attendance and cancellation rates over time and identify changes in high-risk segments.

6. Limitations

Several limitations should be considered when interpreting the analysis:

- The dataset is observational, so relationships identified in the analysis should not automatically be interpreted as causal.
 - Reminder status is associated with different no-show rates, but the analysis does not prove that reminders caused the difference.
 - Some previous-no-show groups contain very few appointments, particularly the groups with four and five previous no-shows.
 - Distance and waiting-time variables contain a small amount of missing data.
 - The analysis identifies patterns and associations but does not independently predict which individual patient will miss an appointment.
 - The dataset is synthetic/anonymised and therefore may not fully represent real-world patient behaviour.
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7. Week 7 Testing Focus

The Week 6 findings provide several areas for further testing in Week 7.

The Data Analytics track should focus on:

1. Testing whether the identified no-show patterns remain consistent across different segments.
2. Monitoring the stability of the validated KPIs.
3. Testing the dashboard calculations and filters for accuracy.
4. Reviewing whether high-risk segments identified in Week 6 remain meaningful.
5. Further investigating the relationship between previous no-shows, reminders and attendance.
6. Checking whether recommendations based on the analysis remain supported by the evidence.
7. Working with other tracks to evaluate how the analytical findings perform when incorporated into the wider HealthConnect solution.

This is consistent with the Week 6 progression toward the Week 7 testing and refinement stage.

8. Week 6 Project Summary

Planned

The main objective for Week 6 was to deepen the Week 5 analysis, validate important KPIs, identify stronger decision-support insights and improve the existing Power BI dashboard.

Completed

The Data Analytics track completed additional analysis of:

- Previous no-show history
- Reminder status
- Appointment type
- Age group
- Booking lead time
- Distance and waiting time

The Week 5 KPIs were also validated.

Main Improvement

The analysis moved from mainly describing appointment outcomes to examining **rates and relationships between different factors and no-show behaviour**.

The Power BI dashboard was also improved to present these findings more clearly for decision-making.

Main Finding

Previous no-show history emerged as one of the strongest indicators associated with current no-show behaviour, with no-show rates increasing from approximately 43.5% among appointments with no previous no-shows to approximately 67.9% among those with three previous no-shows.

Main Business Opportunity

HealthConnect has an opportunity to reduce missed appointments through more targeted patient engagement, particularly for appointments associated with previous no-show behaviour and longer booking lead times.

Remaining Work

Further testing is required to determine whether the identified patterns remain stable and how they perform when incorporated into the wider HealthConnect solution.

Week 7 Focus

The next stage will focus on testing, refinement and validation of the analytical findings and their contribution to the overall HealthConnect project.

9. Conclusion

The Week 6 Data Analytics work strengthened the HealthConnect appointment analysis by moving beyond basic descriptive analysis toward decision-support insights.

The findings confirm that missed appointments remain a significant issue, with a no-show rate of approximately **48.5%**. Previous no-show history showed the clearest relationship with current appointment outcomes, while reminder status, appointment type, age group and booking lead time also showed useful patterns.

The improved Power BI dashboard provides HealthConnect with a practical way to monitor these patterns and support evidence-based decisions.

The analysis should now progress into Week 7 testing and refinement rather than being treated as a final production solution.
